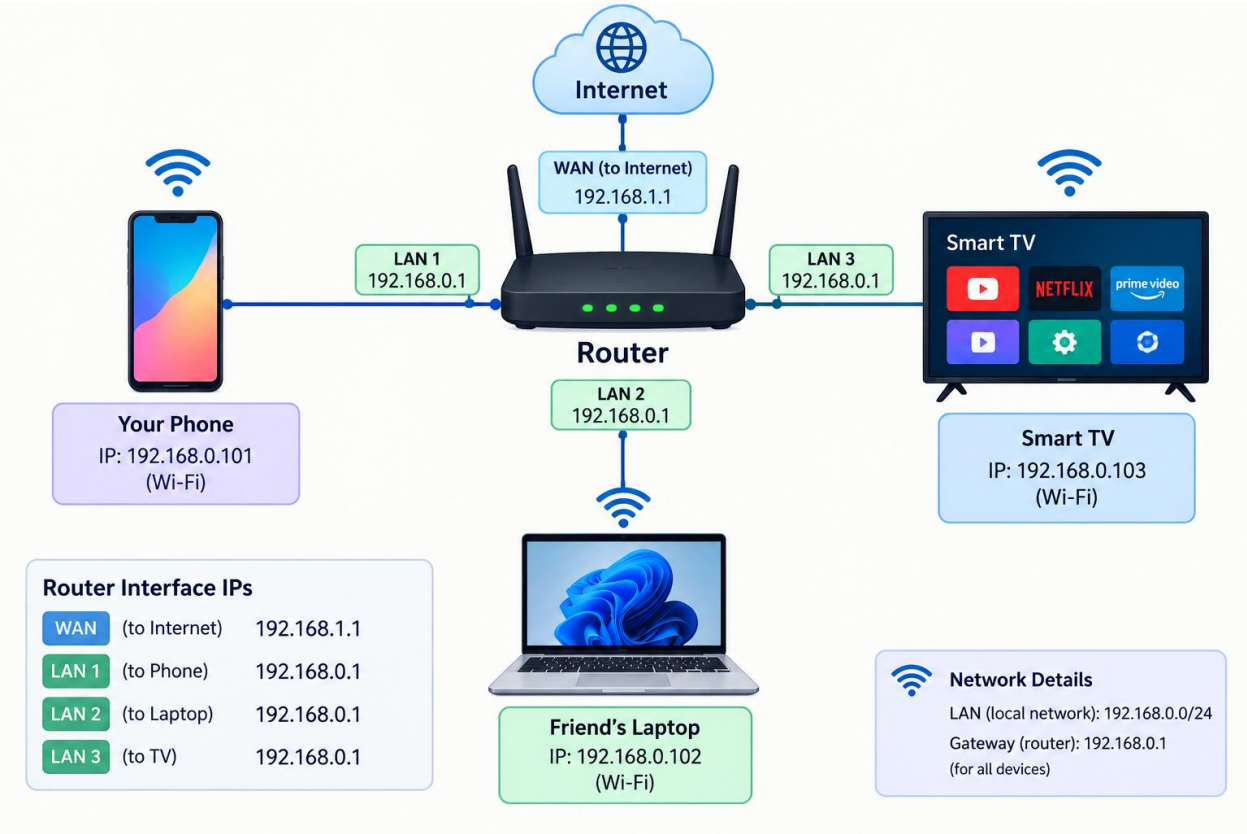
Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



**Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.**

Manual IP Routing – Step-by-Step Explanation

Let's imagine my laptop wants to send a packet to my friend's computer, which is on a different subnet.

1. Network diagram

My Laptop

IP: 192.168.1.10

Subnet: 192.168.1.0/24

Router 1

192.168.1.1 → 10.0.0.1

Packet travels through the network

Router 2

10.0.0.2 → 192.168.2.1

Friend's Computer

IP: 192.168.2.20

Subnet: 192.168.2.0/24

2. Routing tables

Router 1's routing table

Destination network	Next hop
192.168.1.0/24	Directly connected
10.0.0.0/30	Directly connected
192.168.2.0/24	10.0.0.2

Router 2's routing table

Destination network	Next hop
10.0.0.0/30	Directly connected
192.168.2.0/24	Directly connected
192.168.1.0/24	10.0.0.1

3. How the packet travels

1. My laptop creates a packet. The destination IP is `192.168.2.20`, which belongs to a different subnet.
2. The laptop checks its routing table. It sends the packet to its default gateway, Router 1 (`192.168.1.1`).
3. Router 1 checks its routing table. It finds the route to `192.168.2.0/24` through Router 2 (`10.0.0.2`).
4. Router 2 receives the packet. It sees that the destination network is directly connected and forwards the packet to the friend's computer.
5. The friend's computer receives the packet. The packet reaches `192.168.2.20`.

4. How is the routing decision made?

Routers check the destination IP address against their routing tables. They choose the most specific matching route (longest-prefix match). If no route matches and no default route exists, the router drops the packet.

In simple words: My laptop sends the packet to Router 1, Router 1 forwards it to Router 2, and Router 2 delivers it to my friend's computer.

.

Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Routing Table for a Zomato-Style Delivery Network

Imagine Zomato has three branches in different cities: Jaipur, Udaipur, and Delhi. Each branch has its own subnet, and routers help send data between them.

1. Network diagram

Jaipur Branch

Network: 192.168.1.0/24

Main Router

Connects all three branches

Udaipur

192.168.2.0/24

Delhi

192.168.3.0/24

2. Routing table

The following is an example routing table for the Jaipur main router.

Network address	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	Directly connected	G0/0
192.168.2.0	255.255.255.0	10.0.0.2	G0/1
192.168.3.0	255.255.255.0	10.0.0.6	G0/2

Note: These are fictional example addresses. The next-hop addresses represent the neighboring routers for the Udaipur and Delhi branches.

3. How it works

1. A computer at the Jaipur branch sends data to the Udaipur branch.
2. The main router checks the destination network address.
3. It finds `192.168.2.0` in its routing table.
4. It forwards the data to the next hop, `10.0.0.2`, through interface `G0/1`.
5. The Udaipur router forwards the data to the correct computer.

In simple words: A routing table tells the router where to send data, which next router to use, and which interface to send it through.

Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Static Routing vs Dynamic Routing

Static routing means the network administrator sets routes manually. Dynamic routing means routers learn routes automatically using protocols like RIP or OSPF.

1. Static Routing

Example: A small food delivery office connects two computers and one server using fixed routes.

Pros (Advantages)

1. Easy to control: The administrator decides the route manually.
2. More secure: Routes are not automatically shared with other routers.
3. Less resource usage: It uses less CPU and network bandwidth for routing updates.

Cons (Disadvantages)

1. Manual work: Every route must be added by the administrator.
2. No automatic change: If a route fails, someone may need to update it.
3. Difficult for large networks: Managing many branches takes more time.

2. Dynamic Routing (RIP or OSPF)

Example: A Zomato-style food delivery service has branches in Jaipur, Udaipur, and Delhi. Dynamic routing helps routers find alternative paths if one network link fails.

Pros (Advantages)

1. Automatic updates: Routers learn routes from other routers.
2. Handles failures: Routers can find another available route if a link goes down.
3. Good for large networks: It is easier to manage many branches and servers.

Cons (Disadvantages)

1. Uses more resources: Routing updates require CPU and bandwidth.
2. More complex setup: Protocols such as OSPF need proper configuration.
3. Convergence takes time: Routers may need some time to learn about a network change.

3. Comparison table

Feature	Static Routing	Dynamic Routing
Setup	Manual	Automatic learning
Route changes	Manual update	Updates automatically
Resource usage	Low	Usually higher
Best for	Small, stable networks	Large, changing networks
Example	Small shop network	Food delivery service with many branches

Conclusion: Static routing is suitable for small networks with fixed routes. Dynamic routing is better for large services with many branches because it can adapt when network links fail. RIP is generally simpler, while OSPF is better suited to larger, more complex networks.

**Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.**

OSPF Configuration Between Two Routers

OSPF stands for Open Shortest Path First. It is a dynamic routing protocol that helps routers find the best path to send data between networks.

Here is a simple example using Cisco routers.

1. Network diagram

PC 1 — Network 1

192.168.1.10

Router 1

LAN: 192.168.1.1

Link: 10.0.0.1

Router 2

Link: 10.0.0.2

LAN: 192.168.2.1

PC 2 — Network 2

192.168.2.10

2. Router 1 configuration

```
enable
configure terminal
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

Explanation of each line

Configuration line	Meaning in easy language
`enable`	Enters privileged mode.
`configure terminal`	Opens configuration mode.
`router ospf 1`	Starts OSPF process number 1.
`network 192.168.1.0 0.0.0.255 area 0`	Enables OSPF on matching interfaces in the LAN subnet.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the link between the two routers.
`end`	Exits configuration mode.

3. Router 2 configuration

```
enable
configure terminal
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
end
```

Explanation: Router 2 enables OSPF on its own LAN and on the link connecting it to Router 1. Both routers use area 0, allowing them to exchange routing information.

Note: The interface IP addresses must also be configured correctly on both routers before OSPF can work. The `network` commands use wildcard masks, which are the inverse of subnet masks.

4. How to check the configuration

After configuring both routers, use:

```
show ip ospf neighbor
show ip route ospf
```

- `show ip ospf neighbor` — checks whether the routers have discovered each other.
- `show ip route ospf` — displays routes learned through OSPF.

Conclusion: OSPF allows Router 1 and Router 2 to share network information automatically. This helps PC 1 communicate with PC 2 through the routers without manually adding every remote network route.